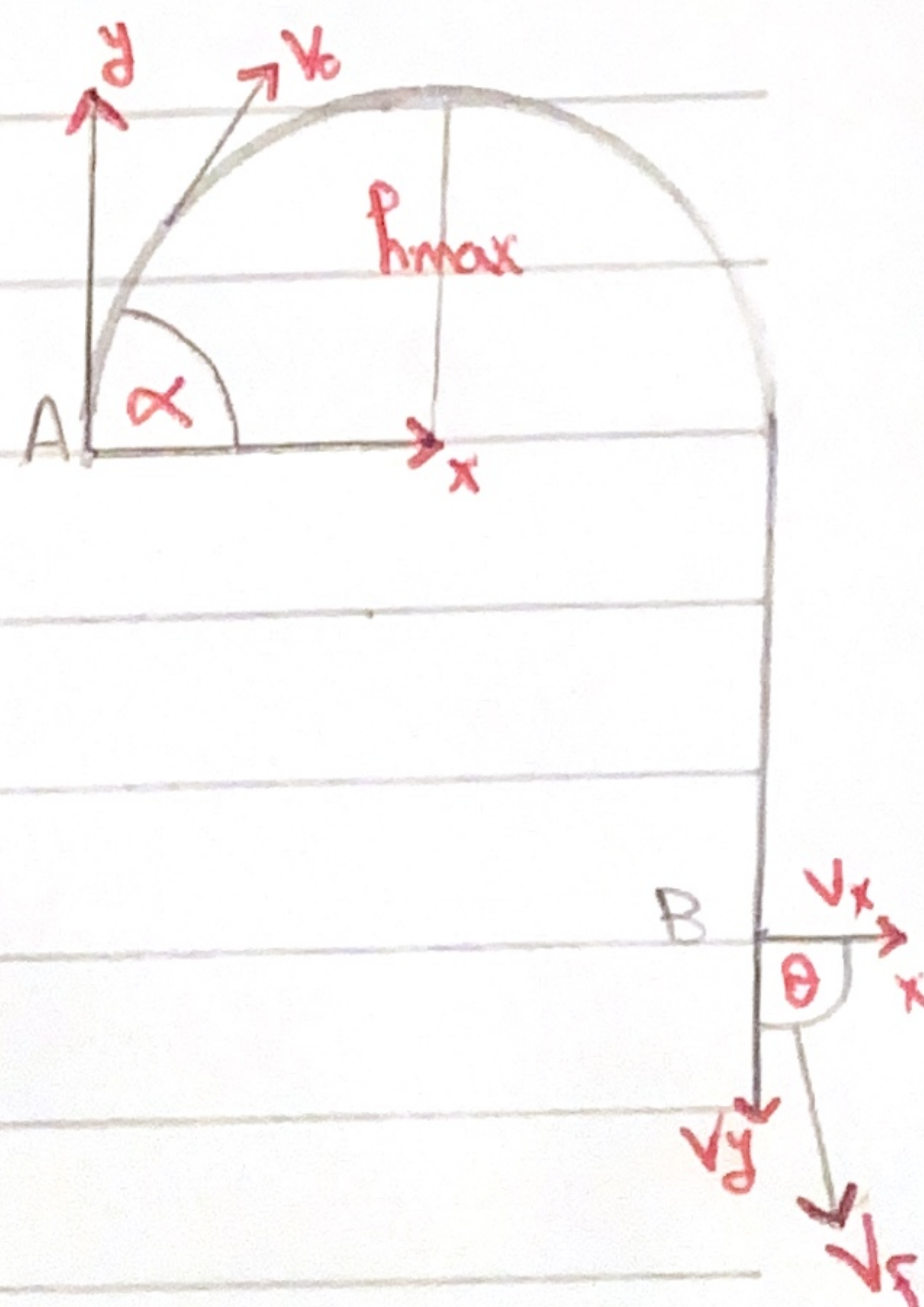


## Lecture 3

- $V_0$  : initial Velocity.
- $\alpha$  : X-axis Angle ( $V_0$ )
- $V_f$  : Final Velocity
- $\theta$  : X-axis Angle ( $V_f$ )



Rules :

- $X = (V_0 \cos \alpha) t$
- $Y = (V_0 \sin \alpha) t - \frac{1}{2} g t^2$

Path eq :

$$Y = X \tan \alpha - \frac{1}{2} g \left( \frac{X^2}{V_0^2 \cos^2 \alpha} \right)$$

final Velocity :

$$V_x = V_0 \cos \alpha$$

$$V_y = V_0 \sin \alpha - g t$$

$$\theta = \tan^{-1} \left( \frac{V_y}{V_x} \right)$$

$$V_{\text{final}} = \sqrt{V_x^2 + V_y^2}$$

Max height :

$$h_{\text{max}} = \frac{V_0^2 \cdot \sin^2 \alpha}{2g}$$